

Design and Implementation of a Real-Time Black Hole Attack Detection System in MANETs using SVM and Live Traffic Analytics

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Abstract:

Mobile Ad-Hoc Networks (MANETs) are known for their susceptibility to routing threats, especially black hole attacks, due to their decentralised and infrastructure-less architecture. Traditional signature-based detection systems often fail to keep pace with the dynamic topology of MANETs. This project implements a high precision Anomaly Detection System based on the Support Vector Machine (SVM) algorithm. The proposed framework utilises a Radial Basis Function (RBF) kernel SVM to classify network behaviour by analyzing high impact traffic features, specifically focusing on Packet Delivery Ratio (PDR), End to End Delay and Throughput. This project integrates a live packet sniffing engine using the Python Scapy library to capture real time network traffic and processes it through a feature extraction pipeline. A core component of this project is the development of a Real Time Security Dashboard built using Streamlit. This interface provides instantaneous visual feedback on network health and triggers automated alerts upon detection of a malicious code. Experimental results show that the SVM based approach achieves a detection accuracy approaching 99.9% and effectively minimizing false positives. The final output provides a lightweight, scalable, and practical security tool for securing modern decentralized wireless networks against routing-layer exploitation attacks.

Reference:

S. Abdallah et al., "Enhancing Mobile Ad Hoc Network Security: An Anomaly Detection Approach Using Support Vector Machine for Black-Hole Attack Detection," *International Journal of Safety and Security Engineering*, vol. 14, no. 4, 2024.

Guide:

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